Lab: Post-Mortem Analysis and Failure Cataloging

Objective

Conduct a structured post-mortem analysis of a testing outcome from your invention project and begin building a failure mode catalog. By the end of this lab, you will have extracted transferable lessons from both failures and successes, and created a learning system that grows across projects.

Prerequisites

- Familiarity with parameter vs. structure learning, post-mortem methods, and failure cataloging from this module's lecture
- At least one testing outcome (positive, negative, or ambiguous) from your invention project
- If you have not yet tested a prototype, you may use a hypothetical scenario based on your design

Materials

- Test records from prior modules (signal collection plan, reasoning worksheets)
- Notebook or digital document for the failure catalog
- Index cards or sticky notes for the abstraction ladder exercise

---

Part 1: Structured Post-Mortem (20 minutes)

Goal: Analyze a specific testing outcome using the five-step post-mortem framework.

Select a testing outcome from your invention project. It can be a failure, a success, or an ambiguous result.

Step 1: Establish the facts (what actually happened, without interpretation)

Write your response here

Step 2: Compare to predictions (what did your generative model predict?)

Aspect	Predicted Outcome	Actual Outcome	Prediction Error (difference)

Write your response here

|

Write your response here

|

Write your response here

|

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

|

Write your response here

|

Write your response here

|

Write your response here

|

Step 3: Attribute causes (for each prediction error, what caused it?)

Prediction Error	Possible Cause 1	Possible Cause 2	Most Likely Cause	Parameter error or Structure error?

Write your response here

|

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

|

Write your response here

|

Write your response here

|

Write your response here

|

Step 4: Extract lessons (what specific updates should be made to your generative model?)

Parameter updates (change values, keep model structure):

Write your response here

Structure updates (add variables, change relationships, remove incorrect assumptions):

Write your response here

Step 5: Assess transferability (are these lessons specific to this prototype or applicable to future projects?)

Write your response here

Part 2: Failure Mode Catalog (15 minutes)

Goal: Create a structured failure catalog entry for each failure observed in your testing.

If your test outcome was a success, imagine three plausible failure modes that could have occurred and catalog those instead.

Field	Entry 1	Entry 2	Entry 3
Date / Prototype Version			

Write your response here

|

Write your response here

|

Write your response here

| | Failure Description (what happened) |

Write your response here

|

Write your response here

|

Write your response here

| | Failure Mode (how it failed) |

Write your response here

|

Write your response here

|

Write your response here

| | Root Cause (why it failed) |

Write your response here

|

Write your response here

|

Write your response here

| | Severity (minor/major/critical) |

Write your response here

|

Write your response here

|

Write your response here

| | Detectability (easy/moderate/hard to detect) |

Write your response here

|

Write your response here

|

Write your response here

| | Lesson Learned |

Write your response here

|

Write your response here

|

Write your response here

| | Countermeasure (how to prevent recurrence) |

Write your response here

|

Write your response here

|

Write your response here

|

Do you see any patterns across your failure entries? Are there common failure modes, common root causes, or common lessons?

Write your response here

Part 3: Success Analysis (10 minutes)

Goal: Apply the same rigor to understanding why something worked.

Select a positive test outcome (or the best-performing aspect of an ambiguous outcome):

What went right?

Write your response here

Did the generative model predict this success? Was the prediction correct for the right reasons?

Write your response here

What factors contributed to the success?

Factor	Was this factor part of your model?	Could this factor have been luck or uncontrolled variables?

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

|

Write your response here

|

Write your response here

|

What aspects of the success does your model NOT explain?

Write your response here

What would happen if you tried to replicate this success in different conditions? Would the same factors still apply?

Write your response here

Part 4: The Abstraction Ladder (10 minutes)

Goal: Practice abstracting specific lessons into transferable principles.

Take the most important lesson from your post-mortem. Climb the abstraction ladder from the specific observation to the meta-principle:

Level	Abstraction	Your Lesson
1. Specific observation	What you saw	

Write your response here

| | 2. Proximate cause | Why it happened in this instance |

Write your response here

| | 3. General mechanism | The underlying principle at work |

Write your response here

| | 4. Design principle | A rule of thumb for future designs |

Write your response here

| | 5. Meta-principle | A general principle about inventing |

Write your response here

|

Which level of abstraction is most useful for your next prototype iteration? Which is most useful for future projects in different domains?

Write your response here

Part 5: Design Your Learning System (10 minutes)

Goal: Create a sustainable system for capturing and applying lessons from every testing cycle.

Design a personal learning system that you will use throughout your invention project:

Where will you record lessons? (Format, tool, location)

Write your response here

When will you conduct post-mortems? (After every test? After each prototype version? Weekly?)

Write your response here

How will you ensure you review past lessons before starting new work? (Checklist? Review ritual? Team discussion?)

Write your response here

How will you organize lessons for cross-project transfer? (Categories? Tags? Abstraction levels?)

Write your response here

Commitment statement: "Before starting each new prototype iteration, I will..."

Write your response here

Discussion and Debrief

1. **Parameter vs. structure**: In your post-mortem, did you find more parameter errors or structure errors? Which type is harder to identify?
2. **Success blind spots**: Was it harder to analyze success or failure? What did the success analysis reveal that you would have missed without structured analysis?
3. **Pattern recognition**: Do you see any patterns in your failure catalog? Are you consistently weak in the same areas?
4. **Abstraction challenge**: Was it difficult to climb the abstraction ladder? At which level did the lesson stop being useful and start being too vague?
5. **Learning commitment**: Do you believe you will actually use the learning system you designed? What is the biggest obstacle to consistent learning discipline?

Write your response here